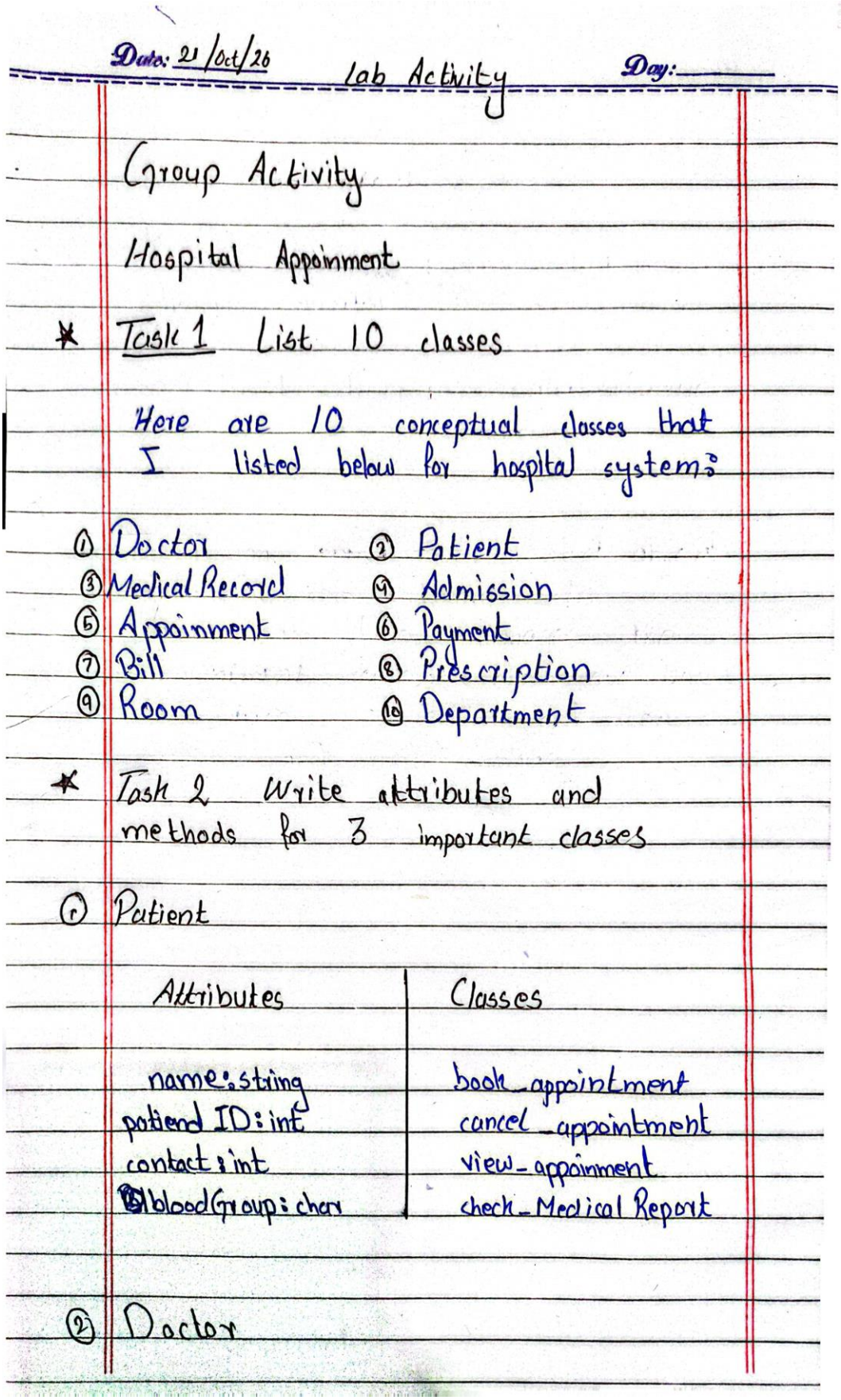
The University Of Azad Jammu and Kashmir, Muzaffarabad

Department of Software Engineering



Student Name	**Muhammad Zain Mughal**
Roll Number	**2025-SE-03**
Program	**BS Software Engineering**
Course Title	**Software Design & Architecture**
Assignment No	**01**
Course Instructor	**Engr. GR Butt**
Semester	**3rd**
Session	**2025–29**
Date	**19-Oct-2026**

Group Activity Performed in Lab



Date: 21/Oct/26 Lab Activity Day:

Group Activity

Hospital Appointment

* Task 1 List 10 classes

Here are 10 conceptual classes that I listed below for hospital system:

1. Doctor
2. Patient
3. Medical Record
4. Admission
5. Appointment
6. Payment
7. Bill
8. Prescription
9. Room
10. Department

* Task 2 Write attributes and methods for 3 important classes

① Patient

Attributes	Classes
name: string patient ID: int contact: int bloodGroup: char	book_appointment cancel_appointment view_appointment check_Medical Report

② Doctor

Attributes	Classes
name: string doctor ID: int Specialization: string Contact Number: int	view Appoinments update Medical Report write prescription

③ Appoinment

Attributes	Methods
appoinment ID: int status: string patient details: string doctor Name: string	schledule appoinments cancel Appoinment check Appoinment

* Task 3: 3 to 4 senteces on abstraction decision.

In this i have only add the important details like patient id, doctor id, appoinment status, appoinment time, etc because these details are required to run a hospital managment system. I ignore the irrelevant information such as skin color, hobbies etc because these

Date: _____ Day: _____

are not important.

* Task 4 Identify one "invented" software object

Appointment Controller → Invented Software Object.

It is an ~~appointment~~ invented object because it help software to work as this manages the appointment request, check availability and time schledule and then confirm and contol booking

